Fourier Transform's Role in Magnetic Resonance

Kyle Arenal, *Member, IEEE,*

***Abstract*—Magnetic Resonance Imaging (MRI) is the non-invasive, multi-step process that spatially maps hydrogen nuclei signals in a body to recreate a three-dimensional image. The challenge in MR is localizing the detected signal. This is achieved by using magnetic field gradients to allow for spatial encoding. Fourier Transform (FT) is applied to analyze the signal and convert this signal into a two-dimensional K-space data matrix. To reconstruct an image, the Fast Fourier Transform (FFT), a discrete Inverse Fourier Transform (IFT), is applied on the K-space matrix. The parameters used in the acquisition of the data for the K-space matrix directly relate to the reconstructed image's Field of View (FOV) and resolution. Under sampling the data leads to aliasing and a warped image. This proves the control the FT has on the final quality of the reconstructed image.**

I. Introduction

MRI manipulates the hydrogen nuclei in the body to create anatomical images. An MRI machine uses strong magnets to redirect the protons and retrieve a signal. The problem with MR is that the signal that is read does not have spatial information. The detected raw signal, known as Free Induction Decay (FID), cannot be mapped to the anatomy of the body alone.

What must take place is the use of an additional magnetic field, known as gradients, to provide the spatial localization. By adding gradients, our raw FID signal now provides a key spatial information for mapping it to the human body. The FT is the mathematical step required to translate the gradient-encoded signal into a set of frequency and phase components. This data is organized into a K-space matrix, which is the bridge between the data from the signals and the final reconstructed image. The quality of our image is highly controlled by how the K-space is sampled, meaning the FT has effect during the processing of data and the reconstruction of it.

The objective is to analyze the role of the FT in the MR imaging sequence. The physics behind MR signal generation and detection are first highlighted to create a base understanding at the foundation of the machine. The process of spatial encoding is then thoroughly explained, highlighting the usage of gradients to perform Slice Selection, Phase Encoding (PE), and Frequency Encoding (FE). After the signal is read and stored in the K-space data matrix, the application of FFT for image reconstruction is discussed. Signal encoding artifacts are explained, providing a direct relationship to the Fourier Transform and the sampling parameters.

II. Fundamentals of MR

MR signal begins with the most abundant resource in the human body, water, and its hydrogen nucleus, or proton. A proton has a property known as 'spin', which is a fundamental property that includes a magnetic effect where the proton behaves like a magnet with north and south poles. Without the effect of an external magnetic field, each proton is angled in various directions. With an abundance of these, each individual magnetic field is cancels another one out and results in a net magnetism of zero.

Introducing a strong external magnetic field, known as B_0, with a fixed direction is the first step in the MR sequence. In clinical use, B_0 is aligned with the length of the body, which is defined as the longitudinal axis or the z-axis. The majority of the protons will tend to align in the same direction. With the slight majority of protons aligned with the B_0 field over those aligned against the field, the magnetic moments do not fully cancel. The sum of these moments create a measurable quantity called the Net Magnetization Vector (M). M is aligned parallel to B_0 and is the source of the detectable MR signal [1].

The protons that align with the B_0 field are not in a fixed alignment. Even though these protons directionally align, they tend to precess, or wobble, around the axis of B_0. This wobble is continuous while the main magnetic field is applied. The angular frequency (ω_0) of this precession is directly proportional to the strength of the main magnetic field (B_0). This relationship between the two is represented in the Larmor Equation:

$$\omega_0 = \gamma B_0 \tag{1}$$

Where ω_0, is the Larmor (precession) frequency (rad/s), γ, the gyromagnetic ratio (42.58 MHz/T for protons), and B_0, the strength of the external magnetic field (T) [2]. This equation defines the specific frequency required to achieve resonance in MR.

III. Excitation and Signal Detection

M being aligned with B_0 is static and undetectable. To generate a measurable signal, we must apply a radio frequency (RF) pulse, B_1, perpendicular to B_0. The B_1 pulse must exactly match the Larmor frequency (ω_0) to cause energy transfer to occur, known as resonance. The pulse itself has a bandwidth, when combined with a magnetic field gradient, allows for slice selection.

The resonant energy transfer excites the protons, causing M to 'tip', moving a portion of the longitudinal magnetization (M_z) into the transverse plane (M_{xy}) [3]. The angle of the tip depends on the RF pulse's strength and duration. Typically a 90-degree RF pulse is applied to move all magnetization from M_z to M_{xy}. When the B_1 pulse is removed, the transverse component (M_{xy}) will precess at the Larmor frequency while its magnitude will begin to decay exponentially. M_{xy} will have an natural decay known as T2 relaxation (or spin-spin relaxation) [3]. In a similar manner, M_z will increase, but at different rates for different tissue types, and be called T1

relaxation (or spin-lattice relaxation) [3]. Both T1 and T2 relaxation are parameters used by MR scanners to generate contrast in the reconstructed images to differentiate tissue types.

From there, the rotating magnetic field will induce a measurable voltage. or electromotive force (emf), in a receiver coil (Faraday's Law) [1, 4]. This raw, oscillating signal that is detected is known as the FID. The FID is measured in a receiver coil as a single composite signal from the entire excited volume, containing no spatial information.

This is the problem with the FID signal, it is a composite signal that provides no spatial information. In MRI, the bulk of the spatial information comes from additional hardware components known as gradients [5]. When a gradient is in effect, we are able to map the nuclear magnetic behavior to a specific region and produce an image of that slice. These mappings provide a spatial arrangement of different chemical environments within it and forms the foundation of MRI. In practice, an MRI gathers many slices and create a three-dimensional view of the internal structure when put together [2].

IV. Spatial Encoding

The challenge of MR imaging is to harmonize the detected signal with the spatial location of its source. As previously stated, the raw FID signal has no inherent spatial data. This is fixed by using the fact that the Larmor frequency can be made to be spatially dependent through applying a non-uniform magnetic field across the sample. Spatial encoding is achieved through the usage of the magnetic field gradients (G_x, G_y and G_z). The gradient field (B_G) is parallel to the static field B_0, with strength varying as a function of position r. The spatially varying magnetic field causes the Larmor frequency (ω_0) to directly map to spatial position, as shown below [6]:

$$\omega(r) = \gamma[B_0 + B_G(r)] \tag{2}$$

For the purposes of this paper, we will review the process under the assumption of a single receiver coil system and 2D Fourier Transform (2DFT). Due to γ, the Larmor frequency is a linear function of the magnetic field. FT now has the ability to separate the signal into its component frequencies. Gradients must be applied so that B can be mapped to positions.

The first sequence in our encoding is to select our slice. When the B_1 pulse is applied, the G_z gradient is applied simultaneously to select a single, thin 2D imaging slice. The thickness of the slice Δz is determined by the specific bandwidth of the B_1 pulse. Due to the G_z gradient, only the resonant frequencies of the protons within this slice are excited.

The y location is encoded in the spins' phase in a separate step called PE. Following slice selection, we briefly turn on the G_y gradient causing protons to precess at different speeds for a short time based on their y-locations. By turning off the gradient, they relax back to their frequency, based on their x position, but are now out of phase with each other. The amount of phase shift is proportional to their y position.

When a magnetic field gradient (G_x) is intentionally applied along the x-axis during the signal readout (the collection of FID), protons experience a different total magnetic field. A collection of protons at position x will each have a unique Larmor frequency [6]. The signal strength (amplitude) from a location directly correlates to the tissue density at a spot x, the basis of image contrast. The FT is then used to convert the raw signal FID into its component frequencies [6]. With each frequency mapping to a unique position x, the FT determines the location for the original location of the signal. This process is known as FE.

The three gradients work in sequence to gather the spatial information required for imaging. PE encodes the y-location into the phase of the signal before the FID is collected, FE encodes the x-location into the frequency of the signal during the signal readout [6]. The 2DFT process is to first excite the slice, apply the G_y gradient to collect the phase shift proportion to their y position, and then apply a gradient G_x on the x-axis to collect data on each proton's frequency. This process is repeated many times, each varying the G_y gradient's strength, until the the K-space data matrix is filled.

For an $N \times N$ matrix, this process must be repeated N times to fill the k-space matrix line by line. Each time this process repeats, G_y is varied in strength to select a single line N and G_x forms the MR signal along this line [4]. Due to the sequence repeating N times, G_y changing its strength, and the Repetition Time (T_R), PE is the most time-consuming step in 2DFT. The 2DFT operates twice, it analyzes the phase shifts between the lines to reconstruct the y-position and one more time to analyze the frequencies in the signal along each line to reconstruct the x-location. Only by acquiring N y-positions will the FT have enough data to fully reconstruct an image.

This whole process between the RF pulse being sent to the receiving resulting FID signal is received is called Echo Time (T_E) [7]. T1-weighted images are made through a short T_E and T_R times, while T2-weighted images are from longer T_E and longer T_R times. T1-weighted images are particularly useful in highlighting anatomy and fat within the body. These tissues have a low T1 time, meaning they recover quickly. T2-weighted images are most beneficial for areas with fluids, as waters and cerebrospinal fluids (CSF) appear bight [7, 8]. A long T_E time allows for T2 decay to occur, which helps the tissues with a long T2 time, like fluids, to appear brighter.

V. K-Space

The relationship between the spatially-encoded signal and the final image is created using the FT within a two-dimensional data matrix known as K-space. As we've mentioned, a single FID does not form an image by itself. It is only one line of this 2D matrix. The K-space matrix is organized by spatial frequency information, not by position, from the raw MRI signal [9]. The center of the matrix refers to the weak PE (G_y) steps, phases closest to and including zero. These depict the contrast of our image and shape of the object. The edges are the strong PE steps, responsible for defining the sharp edges, fine details, and spatial resolution.

A single, bright pixel in the K-space represents a single, specific spatial frequency. When the IFT is applied, it creates a bright, white image [9, 10]. With the application of three

total white dots, an alternating black and white lined image is produced [9, 10].

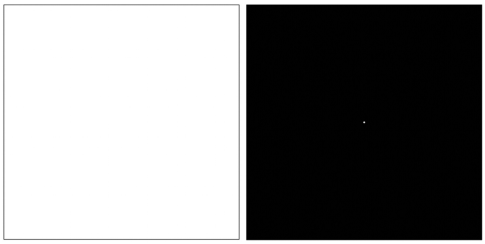

Fig. 1. Left: Image domain showing white canvas. Right: Equivalent k-space containing one bright dot at the origin [9, 10].

Fig. 2. Left: Parallel alternating black and white lines. Right: Equivalent k-space containing three bright pixels aligned vertically [9, 10].

Image quality is determined by the parameters that we use to sample the K-space data matrix. [11] states the FOV and the resolution directly relate to the number of data points acquired in each dimension (G_x and G_y). FOV is determined by the spacing between k_y lines and k_x points, being inversely proportional to our sampling interval (Δk) [12]. To achieve a larger FOV, Δk must be reduced [12]. Resolution is ability to see fine detail in an image. Resolution is determined by the total width of the K-space matrix, which relates to the total number of k_y steps that are sampled. A wider k-space sampling will allow the viewer of the image to see finer details. Under sampling the data, if the k_y steps are too far apart, leads to aliasing and violates the Nyquist criterion [5, 11]. The result is an image incorrectly mapped, causing the signal outside of the FOV to wrap around to the opposite side of an image.

Mitigating aliasing is crucial, and there is a few practical ways you can do it. The simplest solution is to acquire more k_y steps or reduce the Δk spacing, effectively increasing the FOV. Another technique that is used is oversampling, where you gather more data points than required in the PE. This extra data is not necessary, but it's used during image reconstruction to prevent aliasing in our image.

The Signal-to-Noise Ratio (SNR) measures the desired signal from tissue to its unwanted background noise. A higher SNR results in a clearer image where we can now view subtle contrast. A lower SNR means noise is noticeable, making the image appear more grainy. The slice thickness and B_0 have the most effect on SNR. Most clinical MRI involves using $1.5T$ to $3.0T$ when performing a scan, higher values of T up to $7.0T$ generally result in a much higher SNR due to an increased M_0 [13].

FOV can be increased by reducing the T_R, which means that there is a trade-off between field of view, resolution and imaging time [12]. Each parameter cannot be increased without any adverse effect, implying that optimizing the FOV or resolution will cause total imaging time to increase [12].

The k-space trajectory $k(t)$ is determined by the integral, with respect to time, of the gradient function $G(t) = (G_x(t), G_y(t), G_z(t))$, and scaled by γ [14].

$$k(t) = \gamma \int_0^t G(\tau) d\tau \tag{3}$$

The 2DFT utilizes the Cartesian trajectory when sampling. The center of the k-space matrix is often favored by certain trajectories, as the center heavily dictates the image's overall shape, brightness and contrast. The speed the gradient strength can change (S_{max}), or slew rate, and our maximum gradient strength (G_{max}), are the limits that depict our physical and safety constraints [14]. G_{max} and S_{max} determine how quickly we can move from one point in the k-space to another. These are hardware constraints that effect the overall readout time.

Other trajectories such as spiral or radial do not sample the k-space points in a uniform manner. Cartesian works line by line, filling the matrix one spot after the next. Radial and spiral also effectively under sample when acquiring the data. The centers of these regions are dense compared to the boundaries of the matrix [14]. Due to this, FFT does not work immediately and would require another step. This highlights the main reason as to why Cartesian trajectory is very efficient for clinical image reconstruction.

VI. Fast Fourier Transform

The magnetization induces a time dependent signal $S(t)$, which is dependent on the timing and duration of the pulses and the gradients, and $M(r)$ is the spatial distribution of magnetization. [10] Our x and y components are

$$k_x(t) = \gamma G_x t \tag{4}$$

$$k_y(n) = \gamma G_y \tau * n \tag{5}$$

This relationship of the total measured signal is formed in our K-Space Signal Equation:

$$S(k_x, k_y) = \int\int M_{xy}(x, y) e^{-i2\pi(k_x x + k_y y)} dx dy \tag{6}$$

Equation (6) is our measured K-space data $S(k_x, k_y)$, the FT of the image $M_{xy}(x, y)$. To reconstruct the image, we have to apply the IFT to the completed K-space data.

The final step the imaging sequence is to apply the raw data in the K-space ($S(k_x, k_y)$) back into the spatial image ($M_{xy}(x, y)$) using the IFT. Assuming the single, ideal coil, the image $M_{xy}(x, y)$ is obtained by using the 2D IFT:

$$M_{xy}(x, y) = \int\int S(k_x, k_y) e^{i2\pi(k_x x + k_y y)} dk_x dk_y \tag{7}$$

The IFT integral is the mathematical foundation for image reconstruction. In practice, because our K-space data is a finite matrix, we replace the IFT integral with its discrete counterpart and use the Discrete Fourier Transform (DFT). DFT had to be optimized due to a computational speed of $\theta(N^2)$ for an $N*N$ matrix. Computers instead use the optimized algorithm, FFT.

FFT uses a divide and conquer approach, reducing computational time to $\theta(Nlog_2N)$ [6, 13]. DFT also proves to be less memory efficient than FFT, making FFT significantly faster for large data sets. FFT operates under the assumption that N is a multiple of two, allowing for symmetries to occur in the calculation [14].

FFT divides the DFT of length N into two smaller DFTs of length $N/2$ with even and odd subscripts [13]. We are allowed to do this breakdown recursively by the mathematical proof of the double angle formula for trig functions, whose identities we apply under the DFT [13]. If N is a power of 2, we can repeat the division process and reduce our DFT into more equations until we reach DFTs of length 1. We can factor the DFT into smaller lengths if N is not a power of 2, still using the divide and conquer approach [13].

If we had a k-space matrix with a single row of data where $N = 256$, the difference in computational time becomes recognizable. Using the DFT approach of $\theta(N^2)$, it would take 65,536 operations. On the contrary, FFT requires $\theta(Nlog_2(N))$, a much faster 2,048 operations for the same data set. Using this FFT improves our computational speed by 32 times for this size N. This speed increase is a necessity in clinical practice for MRI, where image reconstruction must occur quickly.

Modern clinical scans require larger matrices to increase the resolution and FOV of the image. FFT allows for even larger improvements with these increased matrices when compared to DFT. For matrices of 512×512, 1024×1024 or even larger, the computational difference becomes increasingly pronounced as the matrix grows. Making the improvements made under FFT more recognizable and essential for routine scans.

The efficiency of FFT is not only necessary for standard clinical uses, but also a necessity in advanced applications. Some advanced MRI variants use the spiral, radial, or other non-Cartesian sampling methods for high speed or real-time imaging [17]. Even though the data is non-uniform, the fundamental step of reconstruction still requires us to convert k-space back into the positional image [17]. The data retrieved in this non-uniform trajectories must still be formatted (through gridding) into a FFT compatible format that allows the IFT to operate. Proving that FT converting spatial frequency and position is the foundation for MRIs, regardless of trajectory.

VII. Conclusion

FT is the mathematical operator that elevates MRI from a raw FID signal to a powerful anatomical survey. The utilization is highly critical during both the acquisition of the signal and the image reconstruction phase. The FT translates the spatial encoded signal into the K-space data matrix. The FFT provides the highly efficient image reconstruction necessary for clinical practice. The sampling parameters set during the signal encoding phase directly relate to the produced image, controlling the image's resolution and FOV. Under sampling leads to problems such as aliasing, where images become warped. A strong understanding of the FT is vital to minimize errors during the signal acquisition and to guarantee a quality final image of the anatomical mapping.

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